

Sample Examination

Productions & Logistics

Summer Semester 2021 – Heilbronn Campus

July xx, 2021, Online Examination

Round numbers to the second decimal (e.g., 0.455 to 0.46 or 0.454 to 0.45) unless otherwise stated.

Please note that the language in this exam is not indicative of answers, and (potential) grammar errors do not render an otherwise correct true statement into a false one.

Available time: 120 minutes.

Module: Production

Exercise 1: Multiple Choice Questions (12 Points)

- 1.1** Which of the following is TRUE regarding distributions?
- a) Poisson distribution is a continuous distribution
 - b) Normal distribution function has no inverse function
 - c) Normal distribution is a continuous distribution
 - d) PDF value of any point on any distribution lies between 0 and 1
- 1.2** In an economic order quantity model, Q^* denotes the economic order quantity. Which of the following is correct if the company always orders an exact amount Q^* ?
- a) The company incurs higher annual inventory holding costs compared to annual ordering costs.
 - b) The company incurs lower annual inventory holding costs compared to annual ordering costs.
 - c) The company produces exactly to the demand and works in accordance to lean philosophy.
 - d) The company's annual inventory holding costs and annual ordering costs are same.
- 1.3** Which of the following defines the term capacity utilization?
- a) Ratio of the utilized capacity to the total capacity
 - b) Ratio of the utilized capacity to the available capacity
 - c) Product of the utilized capacity to the total capacity
 - d) Sum of the utilized capacity and the available capacity
- 1.4** Which production system would you choose if you are producing customized apparels, allowing customers to define the material characteristics of the apparel?
- a) Make-to-order
 - b) Engineer-to-order
 - c) Make-to-stock
 - d) Assemble-to-order
- 1.5** Which of the following is TRUE about Order up-to Model?
- a) Multi-period inventory model that considers random demand and fixed lead time
 - b) Two-period inventory model with deterministic demand and random lead time
 - c) Single-period inventory model with random demand and fixed lead time
 - d) Multi-period inventory model with deterministic demand and varying lead time

- 1.6** Which of the following best describes the term “bottleneck” in process analysis?
- a) The activity or task with the least capacity and this activity or task defines the capacity of the entire process
 - b) Average time spent by the flow unit in the process.
 - c) Shortest path from one process to another
 - d) The activity or task with the highest capacity and this activity or task defines the capacity of the entire process
- 1.7** Consider a process that is operating in its steady state. The capacity of the process is 160 units/hour. Every unit enters the process, spends 120 minutes in the process and finally leaves the process. What is the average inventory in the process?
- a) 320 units
 - b) 80 units
 - c) 1.33 units
 - d) 160 units
- 1.8** What value of control limits would you choose when monitoring forecasts if the confidence interval is 95% ($z=1.96$) and the mean squared error (MSE) is 15?
- a) Upper control limit: 7.59 & lower control limit: -7.59
 - b) Upper control limit: 6.96 & lower control limit: -6.96
 - c) Upper control limit: 29.4 & lower control limit: -29.4
 - d) Upper control limit: 0.13 & lower control limit: -0.13
- 1.9** Which of the following is a part of lean transformation?
- a) Allowing back orders in the inventory models.
 - b) Over processing the products.
 - c) Implementing Poka-yoke.
 - d) Produce in batches to reduce the setup time.
- 1.10** What is the optimum batch size of a production system that produces at a rate of 300 units/year, having demand at the rate of 250 units/year, holding cost is €5/unit/year and set up cost is €20/batch?
- a) 109.5 units
 - b) 44.7 units
 - c) 150 units
 - d) 105.3 units

1.11 Which of the following is TRUE regarding random variable?

- a) Random variable is a variable that follows either Poisson or normal distribution
- b) Random variable is a binary variable that is used only in binominal distributions
- c) Random variable is used only in inventory management context to model demand and quantity uncertainties.
- d) Random variable is a function that models the events in the real life

1.12 Which of the following best defines the term “value” in lean terminology?

- a) Value can be anything for which the customer is ready to pay for
- b) Value is the market price that is defined by the company
- c) Value is profit margin of a company
- d) Value is the monetary worth of all the movable and/or immovable assets

Exercise 2: Forecasting (12 points)

Consider the following time series.

Period	Actual demand	Naïve with trend	Exponential Smoothing	Linear Regression
1	97			
2	121			
3	159			
4	127			
5	175			
6	135			
7	143			
MAD				
MSE				

- a) Assume you are in period one and forecast period 2. Then assume you are in period 2 and forecast period 3. Continue until you forecast period 7. Find the following,
- Naïve with trend forecast for Period 5 (1 point)
 - Exponential smoothing forecast for Period 7 ($\alpha = 0.7$ and Initial forecast = 100) (round to 2 decimal places) (1 points)
 - Intercept and slope values of linear regression forecast (round to 2 decimal places) (1 points)
 - Linear regression forecast value for Period 3 (round to 2 decimal places) (1 point)
- b) For the periods 3-7, calculate the MAD and MSE for, (round to 2 decimal places)
- Naïve with trend forecast (1.5 points)
 - Exponential smoothing forecast (1.5 points)
 - Linear regression forecast (1.5 points)
 - Find the best forecasting method (1 points)
- c) If your firm's has high penalty when the forecast value deviates significantly from the actual demand,
- Which forecast evaluation criteria is best suited for your firm? (1 point)
 - Find the optimum value of alpha (α) for your firm using excel solver (round to 2 decimal places) (1.5 point)

Exercise 3: Process Analysis (12 points)

Imagine a process consisting of 5 Workstations that operate 8 hours per day. The factory produces three different products, namely product 1, product 2 and product 3.

Workstations	Number of Employees	Activity time for Product 1 [Min./unit]	Activity time for Product 2 [Min./unit]	Activity time for Product 3 [Min./Unit]
1	2	6	15	0
2	2	5	12	0
3	1	2	7	3
4	1	3	3	3
5	2	10	0	6

Demand information:

Demand for these three products is given in the table below. Note that you might not be able to meet all demand.

Product #	Demand (unit/per day)
1	40
2	45
3	60

- a. What is the daily capacity requirement for resource 1 to produce product 1? Imagine you were to use workstation 1 exclusively for producing product 1 and want to satisfy all demand. How many minutes would workstation 1 be used daily? (2 1 points)
- b. What is the daily capacity utilization of workstation 3 if it produces all three products? Assume, hypothetically, workstation 3 was the bottleneck. What is its capacity utilization when producing all three products? (2 points)
- c. What is the bottleneck workstation in the factory? Explain quantitatively. Select all correct statements (3 4 points)
- d. Assume capacity of each workstation is proportional to the number of employees; one employee is fixed assigned to one workstation. How many employees do you need to add to meet all demand? (2 points)
- e. Assume that product 2 is the most valuable product with a margin that is three times as high as the maximum margin of the other products. Assume you were now to analyze whether to add more employees to resource 3. Which of the following statements best describes the outcome of the optimal analysis? (3 points)
 - a. Up to 2 additional workers should be added to avoid the bottleneck.
 - b. No new workers are required, it is best to prioritize the production of product 3 instead.

- c. Workers should be added and you should also add workers to other resources.
- d. The information provided is not enough to provide a conclusive suggestion.
- d. What is the flow rate (per hour) for each unit flow, assuming that demand must be served in the mix described above (Every 40 units of product 1 there are 45 units of product 2 and 60 units of product 3) (5 points)?

**round numbers to 2 digits after decimal point*

Exercise 4: EOQ and lean (12 points)

A firm named Karl sells e-axle to their customers by operating 4 warehouses with each facing a stable annual demand of 1800 e-axle. Karl incurs an annual inventory holding cost (excluding cost of capital) of €150 per e-axle and Karl's annual cost of capital is 10% of the e-axle value. In addition, processing every order costs Karl €300 in terms of handling, shipping, etc. Karl sells an e-axle at €1750.

- a) What is the total (logistics) cost for Karl (round it to zero decimal places)? (4 points)

If Karl decides to pool their warehouses into one warehouse, assuming that pooling does not incur any extra costs.

- b) What is their cost savings (in %) from warehouse pooling? (2 point)
- c) The annual demand for each of these warehouses increases by 50 percent. What is the new cost savings (in %) from warehouse pooling? (2 points)
- d) Comment the following statement: "When your order quantity is same as the economic order quantity (EOQ) then you incur annual inventory holding costs will be same as your annual order processing (set-up) costs" (1 points)
- e) If Karl produces e-axle at a production rate " p " e-axle/year and if demand consumption starts from the time of production, what would be the effect of in-house production on Karl's annual set-up cost (order processing cost)? (Increases? / decreases? / remains the same?) (1 point)

Assume that you are manufacturing mobile phones in different brand names and all the mobile phones have 90% standard parts and 10% brand specific parts. The demand is randomly distributed and it takes 3 weeks for you to manufacture and assemble a mobile phone. In the entire process, manufacturing parts consume 95% time and final assembly takes only 5% time. The customer expects the mobile phone within 4 days from the ordering date.

- f) Which production system would you choose for your firm? Why? (2 points)

Exercise 5: Practice Problems (12 points)

Consider Kristen Cookies Company as discussed in class and take hints from the lecture slides or case contents (if required) to answer the following,

- a) Briefly state one idea for improving the capacity and justify it (2 points)
- b) Briefly state one idea for improving the demand and justify it (2 points)
- c) Comment the following statement: “Kristen cookies follows make-to-order production system and hence the order lead time cannot be reduced less than an hour” (2 points)

Consider the case OceanCove restaurant as we discussed it in class. Assume that this restaurant is functioning in Covid-19 pandemic period and the local government has imposed a restriction that the restaurant should operate not more than 30% capacity. [Hint: Use the slide contents as reference to answering below questions]

- d) What are the core customer values of OceanCove restaurant? How does the pandemic crisis affects these customer values? (3 points)
- e) Which of the following is the correct business strategy and why? Explain quantitatively (3 points)
 - I. Increasing seating capacity to 160
 - II. Partnering a delivery service company (10% margin will be lost)

Module: Logistics

Exercise 1: Multiple Choice Questions (12 Points)

1. What of the following is NOT TRUE?
 - a. Sharing information to mitigate the bullwhip effect is an easy task.
 - b. We can restructure return policies to eliminate incentives.
 - c. We have to find the root cause of Bullwhip effect to eliminate it.
 - d. Lean management is one of the mitigation strategies against the Bullwhip effect.
2. Which of the following is NOT TRUE about “Every Day Low Pricing”?
 - a. It brings stable demand
 - b. It eliminates strategic buying, decreases volatility and increases efficiency
 - c. It increases strategic buying, decreases volatility and increases efficiency
 - d. It decreases volatility of demand and increases efficiency of supply chains
3. Which of the following is an “objective” in Knapsack problem, if we try to solve it in Excel?
 - a. Maximize the value of the product.
 - b. Minimize the value of the product.
 - c. Maximize the weight of the product.
 - d. Minimize the weight of the product.
4. What of the following is TRUE about Newsvendor, EOQ and Order-up-to models?
 - a. These models consider only physical values
 - b. These models consider only monetary values
 - c. These models consider physical and monetary values
 - d. These models consider only physical values, except EOQ
5. What is TRUE about effect of Corona pandemic on SCF?
 - a. Corona pandemic increased SCF adoption
 - b. Corona pandemic decreased SCF adoption
 - c. Corona pandemic did not have any effect on SCF adoption
 - d. None of the above
6. Which of the following is TRUE about Operations Strategy?
 - a. Strategic decisions are to be aligned with operational decisions
 - b. Differentiation as a strategic advantage happens when low cost positioning is industry wide

- c. Value added in Production & Logistics is value perceived by the suppliers minus costs
 - d. None of the above
- 7. Shortage gaming is one of the causes of bullwhip effect. Please choose from the list below the category to which shortage gaming belongs.
 - a. Information distortion
 - b. Behavioral factors
 - c. Individual incentives
 - d. None of the above
- 8. Consider a release of new Yeezy shoes. As a real fan of Yeezy and Kanye West you place preorders in multiple websites. You want to have the shoes on the day of release to the public and understand that these Yeezy shoes will be in a short supply. So, you place a preorder with multiple retailers; however, you know that you will buy only one and cancel the orders from the others. The question is: To what of the stated concepts below does this example belong?
 - a. Forward Buying due to the information distorting
 - b. Shortage gaming due to the individual incentives
 - c. Shortage gaming due to the behavioral factors
 - d. Reactive & over-reactive ordering due to behavioral factors
- 9. Which of the following are the same in newsvendor inventory model and order-up-to inventory model?
 - a. Ordering strategy
 - b. Inventory management
 - c. Sales Strategy
 - d. Random demand consideration
- 10. Which of the following is TRUE about Prob ($D \leq Q$)?
 - a. Out of stock probability
 - b. In-stock probability
 - c. Service level
 - d. Both (b) & (c)
- 11. What is NOT TRUE about Newsvendor model?
 - a. Single-period inventory model
 - b. Inventories are carried to the next selling season.
 - c. It considers physical units only.
 - d. Used when supply is certain.
- 12. Can we approximate Poisson distribution with a normal distribution?
 - a. Yes, if we know the standard deviation (σ) value, we can approximate Poisson with Normal distribution by using ($\mu = \lambda$)
 - b. No, we cannot approximate Poisson with Normal distribution

- c. Yes, we can approximate by using ($\mu = \lambda$, $\sigma^2 = \lambda$)
- d. No, we cannot approximate a discrete distribution with a continuous distribution

Exercise 2: Inventory Management (12 points)

You are operating a furniture shop in Munich and you are selling desks. You place replenishment orders for desks in a weekly basis and the ordered desks are delivered in three weeks. Space is expensive for your store and you incur annual (52 weeks) holding costs of 30% of desk's value. All stock-outs desks are back-ordered, that is, there are no lost sales and yet you incur opportunity costs (customer dissatisfaction) of 20% of the desk's value per weekly backorder. Assume demand in each week follows a Poisson distribution with mean demand of 16 desks.

(Hint: remember, you are allowed to use Excel but need to briefly explain what you did or add a screenshot!)

- a) Find the optimal service level and express it in percentage with 2 decimal value. (1 point)
- b) Find the optimal order up to level corresponding to optimal service level that you find in (a). (2 points)

Now approximate the weekly demand distribution from Poisson to normal distribution $N(\mu=\lambda, \sigma^2=\lambda)$ (Note: all other values as same as in the above description). Solve the below questions using weekly normal demand distribution.

- c) What is the optimal service level? (1 points)
- d) Find the optimal order up to level for the optimal service level in (c). (2 points)
- e) If the expected lost sales value is 0.02 desk, find the following and express them in 2 decimal places
 - a. Expected back order quantity (1 point)
 - b. Expected left over inventory (2 points)
 - c. Expected Sales (1 point)

Consider a product with newsvendor demand. The demand is uniform and continuously distributed between 150 and 300 units. Company incurs a cost of €2/unit and gains a profit margin of €3/unit.

- f) Find the optimum number of units that are to be ordered (2 points)

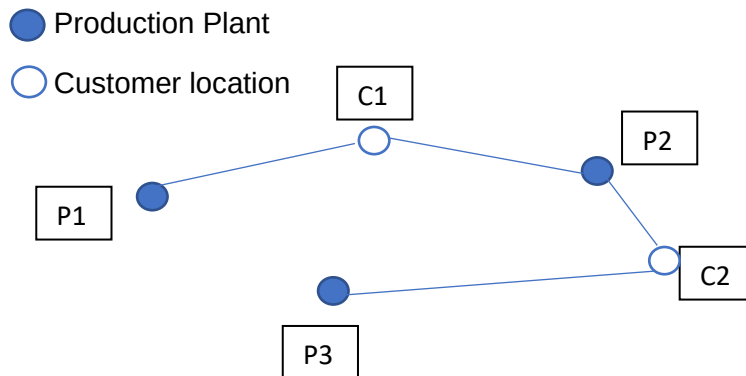
Exercise 3: Supply Chain Coordination (12 points)

Imagine retailer “Fancy People Only,” which usually sells 200 packs of Almond Milk per week, assumes that people drinking such milk increased suddenly as the most famous TikTokker posted a “drink only Almond Milk” challenge. In the following week, the retailer ordered 350 packs of almond milk from the “Fancy Milk Maker” manufacturer instead of ordering the usual size, that is, 200 packs. The manufacturer is happy with such a big order and it in turn ordered even more almonds from its supplier.

- a. By how many percentage points does the demand increase? (75%)
- b. Following the logic of the Bullwhip effect in supply chains, which of the following stages of the supply chain do you expect to be affected ultimately most by this increase? (2 points)
- c. As such action leads to the distorted picture of customer demand for the manufacturer, what problems might it lead to in the future? Select all correct statements below and remember that a statement is only correct if it is entirely true (i.e., including the explanation) (6 points).
- d. Please define what caused the Bullwhip effect in this case and put it into one of the categories. Explain the cause with an example and explain the category. (4 points)

Exercise 4: Supply Chain Network Design (12 points)

There are two customers (C1, C2) and you can open three plants to serve each customer. Cost of opening each plant is 4, 5, 7, respectively. The following graph shows which plant can serve which customer. For example, P1 can serve C1.



- Formulate the problem as a mixed integer linear problem and introduce the decision variables (3 points).
- Solve this problem applying the location covering problem heuristic (4 points).
- Do you think your solution is optimal? What are the other ways to find an optimal solution OR how can you proof check your solution (2 points).

We want to pack out backpack. The maximum weight allowed is 12 kg. Now, we have to choose between sleeping bag (8 kg), Torch (1kg), Water bottle (2kg), Rope (3kg) and a laptop (6kg). Personal value of each item is 12, 5, 8, 2, 6 respectively. Which products should we pack? (3 points) –write down the product names and their total value and weight.

Exercise 5: Practice Problems (12 points)

Imagine you are a huge retail company in the United States, and you have more than 500 suppliers. You can only pay after the goods were delivered, and you also try to prolong your payments to suppliers. You know that your retailers need money right after the delivery of goods as they need to continue their operations, but you cannot pay them immediately at all.

- a. State what might be a potential solution to satisfy needs of both suppliers and a buyer (recall from the lecture). (1 points)
- b. Which parties are involved in this solution (1 point)?
- c. How do all parties benefit from this solution (2 points)?
- d. What kind of difficulties buyers might face by implementing solution you have stated (2 points)?

Consider the HP case:

- a. Describe three key performance indicators relevant for HP and explain why they are important. (3 points)
- b. Please state and explain the main issues HP was facing (3 points).
- c. If the HP case was taking place today. What two technologies would you recommend that firm to use and why would they help to solve the key issue?